TI TIDA-060029 analog front end

Matched RG/RF: 100R / 5k / 100k

File: afe.kicad_sch

DDS excitation and reconstruction

100Hz-100kHz small-signal excitation

File: source.kicad_sch

Simultaneous voltage / current acquisition

Two channels / one conversion instant

File: adc.kicad_sch

USB, RP2040 and boot / debug

USB control, firmware and host calibration

File: digital.kicad_sch

Power: 5V input, forced PWM, quiet analog rails

External 5V → quiet +/-12V

File: power.kicad_sch

Engineering prototype. One PCB. Isolated, discharged DUT only.

Temper

Sheet: /
File: temper-lcr.kicad_sch

Title: Temper LCR — TIDA-060029 based

Size: A3	Date: 2026-10-20	**Rev: D**
KiCad E.D.A. 10.0.4		Id: 6/6

U6A source buffer; U6B HI sense; U7A transimpedance; U7B LO sense.
Fit ONE MATCHED pair: JP1+JP4=100R; JP2+JP5=5k; JP3+JP6=100k.
Power OFF to change ranges. One local C104=100pF compensates all ranges.
No BUF634A: maximum source current approximately 18mA. Isolated, discharged DUT only.

TI source and range network

JP1
R40
RG 100R
JP2
100R 0.1%
R41
RG 5k
JP3
5k 0.1%
R42
RG 100k
100k 0.1%
U6A
EXCITE
OUT
OPA2810IDGKR
SOURCE_BUF

TI transimpedance topology; Rev C compensation
C104=100pF directly across U7.1 and U7.2 in all ranges

R43
JP4
100R 0.1%
R44
RF 100R
JP5
5k 0.1%
R45
RF 5k
JP6
100k 0.1%
RF 100k
C104
100pF C0G 5%
U7A
GND
OUT
OPA2810IDGKR
TIA_OUT

Rev D: R94 22R 0.1% between J3 LO and the TIA summing node.
LOW follower senses J3 LO; host uses Zf = R94 + RF||C104.

J3
DUT_HI
DUT_LO
DUT HI / LO
R94
22R 0.1%
TIA_SUM

J3: connect external, isolated DUT between HI and LO.
No DUT component is fitted on this PCB.

U6B
HIGH_BUF
OPA2810IDGKR

U7B
LOW_BUF
OPA2810IDGKR

Independent HI / LO voltage followers

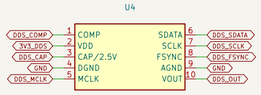
U6C
C42
100nF 50V
C44
1uF 50V
C50
10nF 50V
C43
100nF 50V
C45
1uF 50V
C51
10nF 50V
OPA2810IDGKR

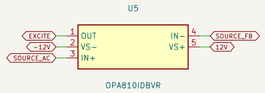
U7C
C46
100nF 50V
C48
1uF 50V
C52
10nF 50V
C47
100nF 50V
C49
1uF 50V
C53
10nF 50V
OPA2810IDGKR

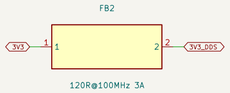
Retain TI 100nF + 1uF bypass per rail; added 10nF at each supply pin.
PCB plane windows and short feedback routes are specified in reference-conformance.md.

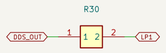
Temper	
Sheet: /TI TIDA-060029 analog front end/	
File: afe.kicad_sch	
Title: TI TIDA-060029 analog front end	
Size: A2	Date: 2026-10-20
KiCad E.D.A. 10.0.4	Rev: D
	Id: 2/6

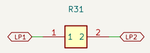
24MHz DDS clock from RP2040 120MHz / 5: sine output only.
Two 330R/1nF RC stages, AC coupling, OPA810 noninverting gain 6.
This block is added to the TI reference. Measure actual amplitude and distortion.

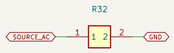


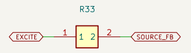


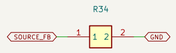














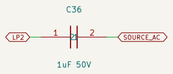




















Global net names connect functional sheets. See TI conformance matrix.

Temper

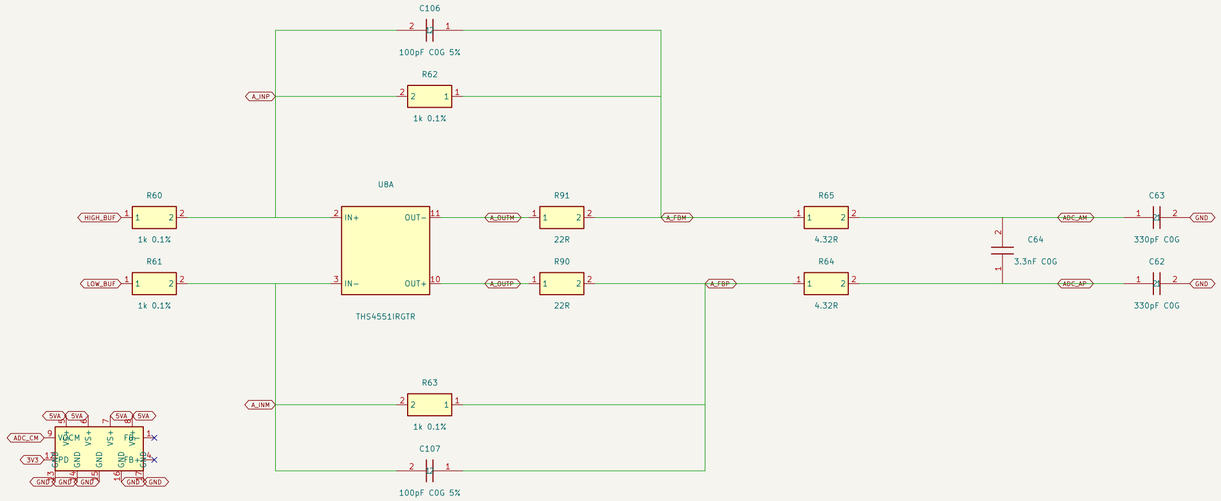
Sheet: /DDS excitation and reconstruction/
File: source.kicad_sch

Title: DDS excitation and reconstruction

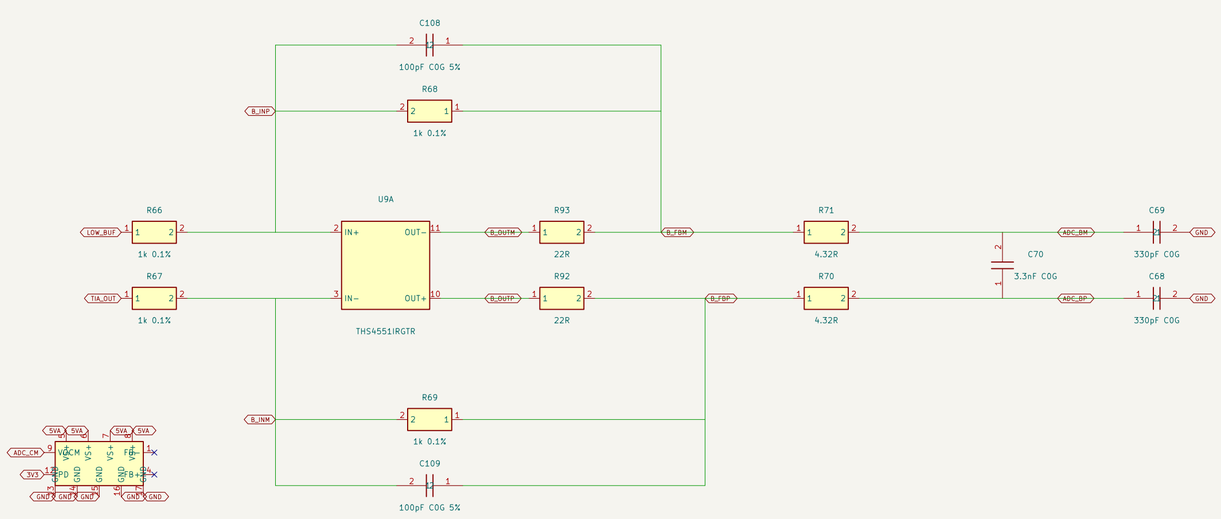
Size: A2	Date: 2026-10-20	**Rev: D**
KiCad E.D.A. 10.0.4		Id: 3/6

Channel A = HIGH_BUF – LOW_BUF; channel B = LOW_BUF – TIA_OUT.
Z = [RF / (1+j*2*pi*f*RF*CF)] * A/B. CF=100pF on every range; high-range pole=15.9kHz.
ADS9224REVM p26 topology: 22R inside feedback, 100pF || 1k, then 4.32R to ADC.
REFby2 drives VOCM. REFOUT is decoupled only. Each REFP uses 22uF.

U8: feedback senses the nodes AFTER 22R, BEFORE 4.32R.



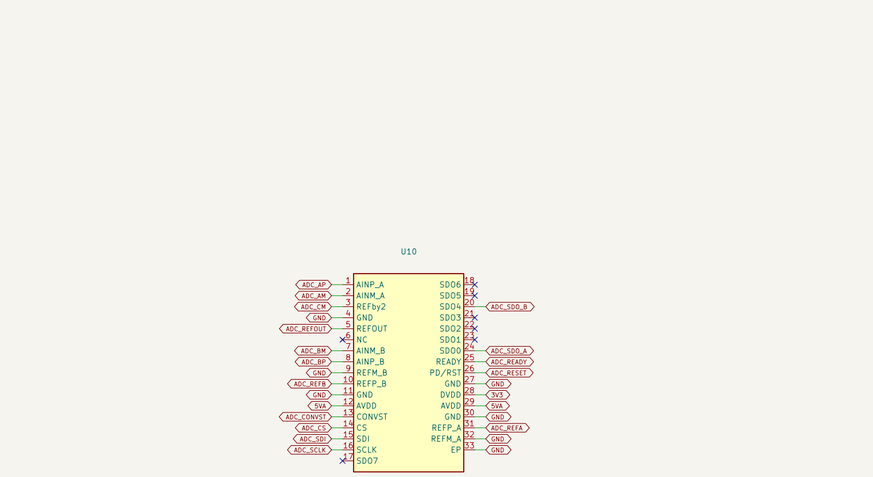
U9: feedback senses the nodes AFTER 22R, BEFORE 4.32R.



U10

ADS9224RIRHBR

Reference and supply bypass. REFby2 directly drives both VOCM pins;
C74=1uF. EVM per-driver 100R / 100nF VOCM filters are omitted.
RGT internal FB pads are NC; external feedback uses main OUT pads.

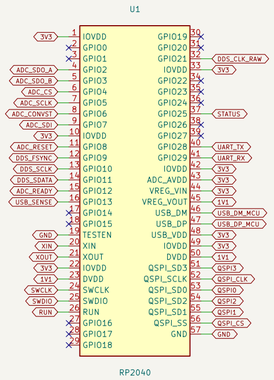


Temper
Sheet: /Simultaneous voltage / current acquisition/
File: adc.kicad_sch

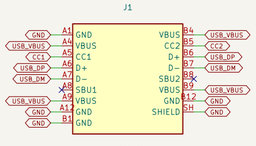
Title: Simultaneous voltage / current acquisition

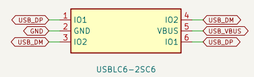
Size: A2	Date: 2026-10-20	**Rev: D**
KiCad E.D.A. 10.0.4		Id: 4/6

External 5V input powers the instrument. USB VBUS is sensed only; no backfeed.
Native USB CDC; 120MHz MCU; dual-lane ADC acquisition at up to 1MSPS.
Hold BOOTSEL while resetting for ROM USB firmware loading. SWD and UART provided.

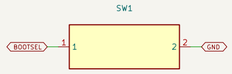


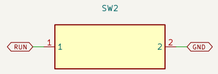


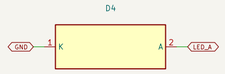


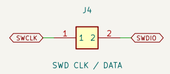


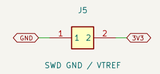


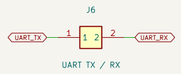


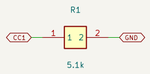


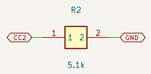


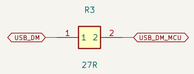


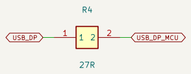


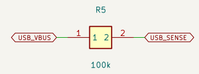


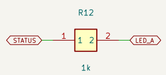


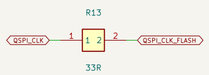






























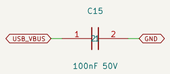












Global net names connect functional sheets. See TI conformance matrix.

Temper

Sheet: /USB, RP2040 and boot / debug/
File: digital.kicad_sch

Title: USB, RP2040 and boot / debug

Size: A2	Date: 2026-10-20	**Rev: D**
KiCad E.D.A. 10.0.4		Id: 5/6

Input: REGULATED 5V +/−5%, 1A, center positive. USB does not supply this board.
TPS65131 PSP/PSN grounded: forced PWM 1.25−1.5MHz. RAW_P ~14.31V: RAW_N ~−14.39V.
TPS7A49 ~+11.97V: TPS7A30 ~−12.00V. Fixed output values are nominal.
Do not substitute capacitor voltage/package without checking DC−bias derating.

U3
5V GND 5V — VIN GND EN — NC VOUT — 3V3
3.3V AP2112K

U11
12V REGP_FB GND LDOIN_P — OUT FB NC GND EN — NR/SS DNC IN EP — REGP_NR GND LDOIN_P GND
TPS7A4901DGNR

U12
−12V REGN_FB GND LDOIN_N — OUT FB NC GND EN — NR/SS DNC IN EP — REGN_NR GND LDOIN_N GND
TPS7A3001DGNR

U13
SW_P GND GND CONV_VIN 5V 5V — INP PGND PGND VIN INN BSW ENP PSP ENN PSN NC OUTN — OUTN VNEG FBN VREF CN AGND NC CP FBP VPOS INP EP — SW_N RAW_N CONV_FBN CONV_REF CONV_CN GND CONV_CP CONV_FBP RAW_P SW_P GND
3V3 GND 3V3 GND SW_N
TPS65131RGER

Q1
GND 5V — G S — D — 5VIN
AO3401A

D2
RAW_P — K — A — SW_P
SS14

D3
SW_N — K — A — RAW_N
SS14

L1
5V — 1 — 2 — SW_P
4.7uH 3.8A shielded

L2
SW_N — 1 — 2 — GND
4.7uH 3.8A shielded

FB1
5V — 1 — 2 — 5VA
120R@100MHz 3A

J2
5VIN GND — TIP SLEEVE — SW
5V 1A CENTER+

R95
RAW_P — 1 2 — LDOIN_P
1R

R96
RAW_N — 1 2 — LDOIN_N
1R

R80
5V — 1 2 — CONV_VIN
100R

R81
RAW_P — 1 2 — CONV_FBP
1.62M

R82
CONV_FBP — 1 2 — GND
150k

R83
RAW_N — 1 2 — CONV_FBN
1.74M

R84
CONV_FBN — 1 2 — CONV_REF
150k

R85
12V — 1 2 — REGP_FB
91k 0.1%

R86
REGP_FB — 1 2 — GND
10k 0.1%

R87
−12V — 1 2 — REGN_FB
91k 0.1%

R88
REGN_FB — 1 2 — REGN_FB
1k 0.1%

R89
REGN_FB — 1 2 — GND
10k 0.1%

C16
5V — 1 2 — GND
1uF 50V

C17
3V3 — 1 2 — GND
1uF 50V

C18
3V3 — 1 2 — GND
10uF 50V

C80
5V — 1 2 — GND
10uF 50V

C81
CONV_VIN — 1 2 — GND
100nF 50V

C82
5V — 1 2 — GND
10uF 50V

C83
5V — 1 2 — GND
100nF 50V

C84
CONV_REF — 1 2 — GND
220nF 50V

C85
CONV_CN — 1 2 — GND
4.7nF 50V

C86
CONV_CP — 1 2 — GND
10nF 50V

C87
RAW_P — 1 2 — CONV_FBP
4.3pF C0G

C88
RAW_N — 1 2 — CONV_FBN
4.3pF C0G

C89
RAW_P — 1 2 — GND
10uF 50V

C90
RAW_P — 1 2 — GND
10uF 50V

C91
RAW_N — 1 2 — GND
10uF 50V

C92
RAW_N — 1 2 — GND
10uF 50V

C93
REGP_NR — 1 2 — GND
10nF 50V

C94
LDOIN_P — 1 2 — GND
10uF 50V

C95
12V — 1 2 — GND
10uF 50V

C96
12V — 1 2 — REGP_FB
10nF 50V

C97
LDOIN_P — 1 2 — GND
100nF 50V

C98
REGN_NR — 1 2 — GND
10nF 50V

C99
LDOIN_N — 1 2 — GND
10uF 50V

C100
−12V — 1 2 — GND
10uF 50V

C101
−12V — 1 2 — REGN_FB
10nF 50V

C102
LDOIN_N — 1 2 — GND
100nF 50V

C103
5VA — 1 2 — GND
22uF 25V

PWR_FLAG GND
PWR_FLAG 5VIN
PWR_FLAG 5V
PWR_FLAG 5VA
PWR_FLAG 3V3_DDS
PWR_FLAG RAW_P
PWR_FLAG RAW_N
PWR_FLAG CONV_VIN
PWR_FLAG LDOIN_P

Global net names connect functional sheets. See TI conformance matrix.

Temper	
Sheet: /Power: 5V input, forced PWM, quiet analog rails/	
File: power.kicad_sch	
Title: Power: 5V input, forced PWM, quiet analog rails	
Size: A2	Date: 2026−10−20 — **Rev: D**
KiCad E.D.A. 10.0.4	Id: 6/6